METAL SOLUTIONS

EOS Copper Cu

Material Data Sheet

EOS COPPER CU

High purity copper to reach good electrical and thermal conductivity. Suitable for a wide variety of applications.

MAIN CHARACTERISTICS

- High purity copper
- Good electrical and heat conductivity
- Process developed to achieve best possible conductivity using the EOS M 290

TYPICAL APPLICATIONS

- Heat exchangers
- Electronics
- Variety of industry applications requiring good conductivity

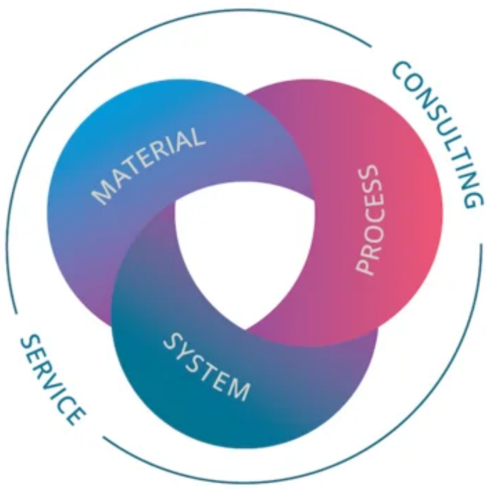
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards



POWDER PROPERTIES

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
Cu	Balance	
O	-	0.35
-	-	0.5

Powder Particle Size

Generic Particle Size Distribution	15 - 45 µm

HEAT TREATMENT

Description

Copper can be heat treated to reach different mechanical properties and conductivity values

Steps

Hold 1 h at ~ 1000 °C in argon atmosphere, slow cooling with argon

PROCESS DATA SHEET

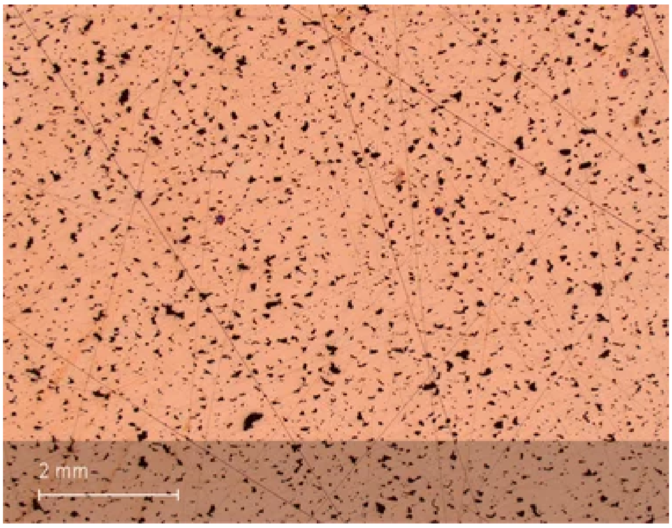
EOS Copper Cu for EOS M 290 | 20 µm

EOS M 290 - 20 µm - TRL 3

System Setup	EOS M 290
EOS Material set	Cu_020_CoreM291_100
Software Requirements	EOSPRINT 2.6 or higher
Recoater Blade	HSS (High Speed Steel)
Nozzle	EOS Grid Nozzle
Inert gas	Argon
Sieve	63 µm

Additional Information	
Layer Thickness	20 µm
Volume Rate	1.7 mm^3/s

Chemical and Physical Properties of Parts



Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	20 µm	< 5 %* %	-

Mechanical Properties

Mechanical Properties As Manufactured

EN ISO 6892-1 Room Temperature | 20 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	180	200	5	-	-	-
Horizontal	180	200	5	-	-	-

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature | 20 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	140	190	20	-	-	-
Horizontal	140	190	20	-	-	-

Copper can be heat treated to reach different mechanical properties and conductivity values.
Properties in the table have been achieved with following heat-treatment:

Hold 1 h at ~ 1,000 °C in argon atmosphere, slow cooling with argon

Electrical Conductivity

ASTM E1004-17	Orientation	Typical Electrical Conductivity [%IACS]
As Manufactured | 20 µm |	Vertical	80
As Manufactured | 20 µm |	Horizontal	80

ASTM E1004-17	Orientation	Typical Electrical Conductivity [%IACS]
Heat Treated | 20 µm |	Vertical	90
Heat Treated | 20 µm |	Horizontal	90

HEADQUARTERS

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